

ROBO SOCCER

DISQUALIFICATION & SAFETY GUIDELINES

INTRODUCTION

IN ROBOSOCCER, TWO ROBOTS WILL GO HEAD-TO-HEAD IN A SOCCER MATCH, PUSHING OR KICKING A TENNIS BALL TO SCORE GOALS AGAINST EACH OTHER.

THE ROBOTS, DESIGNED WITH WEDGES OR OTHER SURFACES TO PUSH THE BALL, COMPETE ON A WOODEN SOCCER FIELD WITH GOALPOSTS.

THE TEAM THAT SCORES THE HIGHEST GOALS WITHIN THE 3-MINUTE ROUND TIME LIMIT ADVANCES TO THE NEXT STAGE.

ROBOT

- 1 **WEIGHT LIMIT:** 3 KG (NO TOLERANCE)
- 2 **BOT DIMENSIONS:** 30 CM * 30 CM * 30 CM (INCLUDING TIRES)
- 3 **CONTROL:** MANUAL CONTROL (WIRED & WIRELESS ALLOWED)
- 4 **POWER SUPPLY:** BATTERY OPERATED (MAXIMUM 24V) FOR WIRELESS & AC SUPPLY OF 220V AC WILL BE PROVIDED FOR WIRED BOTS (TEAMS WILL NEED TO BRING THEIR OWN SMPS)
- 5 **CONTROL:** ONLY ONE PARTICIPANT PER ROBOT
- 6 **GRIPPING:** NO GRIPPING MECHANISMS ALLOWED; ROBOT CAN ONLY PUSH THE BALL

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RULES & REGULATIONS

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- 1 ONLY ONE PARTICIPANT SHOULD CONTROL THE ROBOT THROUGHOUT THE MATCH; NO PLAYER EXCHANGES ARE ALLOWED
- 2 HUMAN INTERFERENCE, SUCH AS TOUCHING THE ROBOT, IS STRICTLY PROHIBITED DURING THE GAME
- 3 ROBOTS CAN PUSH OR HIT THE BALL, BUT THEY CANNOT WITHHOLD OR GRAB IT
- 4 IF THE GAME NEEDS TO BE PAUSED, PENALTIES WILL APPLY IF THE PAUSE LASTS LONGER THAN ONE MINUTE
- 5 IN CASE OF ACCIDENTAL GRIPPING OF AN OPPONENT'S ROBOT, MATCH WILL BE PAUSED, AND THE TIMER WILL BE HALTED UNTIL THE ISSUE IS RESOLVED. MATCH RESUMES FROM SAME POSITION ONCE RESOLVED
- 6 BALLS STAYING IN THE CORNER OF THE ARENA FOR 10 SECONDS WILL BE REPOSITIONED TO THE CENTER AFTER A PAUSE OF 3 SECONDS

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MATCHES, MARKING & PENALTIES

MATCHES

MATCH FORMAT, ROUNDS & MARKING

- 1** **FORMAT:** TEAMS COMPETE ONE-ON-ONE. THE ROBOT THAT SCORES THE HIGHEST NUMBER OF GOALS WITHIN THE TIME LIMIT WINS THE ROUND AND ADVANCES TO THE NEXT ROUND.
- 2** **ROUND TIME:** 3 MINUTES PER ROUND
- 3** **MARKING:** THE ROBOT THAT SCORES THE MOST GOALS WINS

PENALTIES

- 1** **TIME PENALTY:** IF THE MATCH EXCEEDS A 1-MINUTE PAUSE (CALLED BY A TEAM), PENALTIES WILL BE APPLIED.
- 2** **DISQUALIFICATION:** FAILURE TO COMPLY WITH THE ROBOT SPECIFICATIONS (E.G., USING ILLEGAL MECHANISMS) WILL LEAD TO IMMEDIATE DISQUALIFICATION.

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DISQUALIFICATION & SAFETY GUIDELINES

DISQUALIFICATION

- 1 ROBOTS THAT UTILIZE ILLEGAL MECHANISMS, SUCH AS GRIPPING DEVICES WILL BE DISQUALIFIED IMMEDIATELY.
- 2 ANY FAILURE TO ADHERE TO THE ROBOT SIZE AND POWER SUPPLY SPECIFICATIONS MAY LEAD TO DISQUALIFICATION.

SAFETY

- 1 PARTICIPANTS MUST ENSURE THAT THEIR ROBOTS ARE DESIGNED SAFELY TO AVOID ANY DAMAGE TO THE ARENA OR HARM TO OTHER PARTICIPANTS.
- 2 ROBOTS EMITTING INFRARED LIGHT OR USING ELECTRICAL WEAPONS THAT CAUSE INTERFERENCE WILL NOT BE ALLOWED.
- 3 ANY ACCIDENTAL ROBOT ENTANGLEMENT WILL RESULT IN A TEMPORARY PAUSE, BUT THE SAFETY OF PARTICIPANTS WILL ALWAYS BE PRIORITIZED.